

Experiment 6: The most frequent letter of the ciphertext

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>


int main()
{
    char cipher[100];

    int i, c, p;

    int a = 3, b = 15;

    int inverse = 9;


    printf("Enter ciphertext: ");

    scanf(" %[^\\n]", cipher);


    printf("Decrypted text: ");

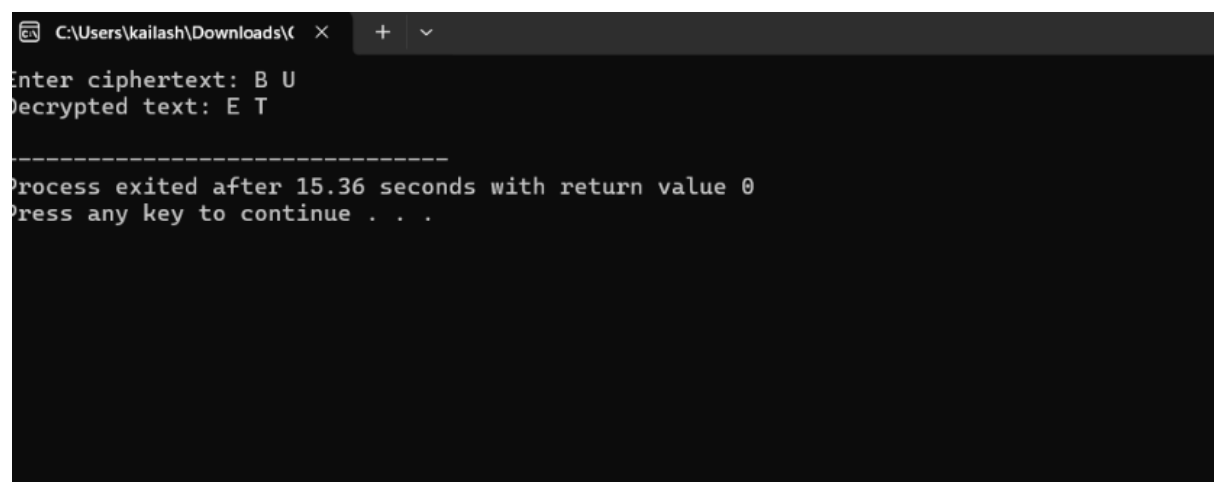

    for (i = 0; cipher[i] != '\\0'; i++)
    {
        if (isalpha(cipher[i]))
        {
            c = toupper(cipher[i]) - 'A';

            p = (inverse * (c - b + 26)) % 26;


            printf("%c", p + 'A');
        }
        else
        {
```

```
        printf("%c", cipher[i]);  
    }  
}  
  
printf("\n");  
return 0;  
}
```

Output:



```
Enter ciphertext: B U  
Decrypted text: E T  
-----  
Process exited after 15.36 seconds with return value 0  
Press any key to continue . . .
```